2024-Spring (Pre-Board) Python

1.a) Define Cloud Computing. List and explain application of Cloud Computing.

sol:- Cloud Computing is a technology that uses remote servers on the internet to store, manage and access data online rather than the local drivers.

Cloud computing has a wide range of applications across different domains. Some of the applications of cloud computing are as follows:-

(i) Backup and Recovery of data:

→ Cloud platforms like Google Drive, Dropbox, and Amazon S3 allow users and businesses to store and backup their data online.

(ii) Web Hosting :

→ Websites and web applications are hosted on cloud servers using services like AWS, Microsoft Azure, and Google Cloud.

(iii) Software as a Service (SaaS):

→ Cloud-based software like Gmail, Ms-office 365, Salesforce and zoom allow users to access application over the internet.

(iv) Disaster Recovery:

→ Organizations use cloud-based disaster recovery solutions to replicate their data and infrastructure.

(v) Development and Testing:

→ Developers use cloud environments for building, testing and deploying applications.

(vi) E-Commerce:

→ Online retailers use cloud platforms to manage their websites, customer databases, inventory and payment systems.

(vii) Education and E-Learning:

→ Institutions and educators use cloud services like Google classroom, Microsoft Teams for remote learning.

(viii) Healthcare:

→ Hospitals use cloud platforms for managing electronic health records (EHRs), telemedicine and data sharing.

1.b) List and explain different cloud services.

Sol The cloud services are typically categorized into three main ~~com~~ services models. They are:-

(i) Software as a Service (Saas).
(ii) Platform as a Service (Paas)
(iii) Infrastructure as a Service (Iaas)

(*) Software as a Service (Saas):

→ In Saas, Cloud delivers software over the internet on a subscription basis, eliminating the need for installation and maintenance.
→ Saas provides web software and applications to complete business tasks.
→ Saas provides the services to the end-users.
→ Saas is used by end-users.
→ Saas provides : Infrastructure + Platform + Software.
→ Examples: Email, MS-365, zoom, Salesforce etc.

(ii) Platform as a Service (Paas):

→ In Paas, cloud offers a platform for developer to build, test and deploy applications.

→ Paas provides virtual platforms and tools to create, test and deploy applications.

→ Paas provides runtime environments and deployment tools for applications.

→ Paas is used by developers.

→ Paas provides: Infrastructure + Platform.

→ Examples: Heroku, Google App Engine, ~~etc~~; Azure App Service etc.

(iii) Infrastructure as a Service (Iaas):

→ In Iaas, cloud provides virtualized computing resources over the internet on demand.

→ Iaas provides a virtual data center to store information.

→ Iaas provides access to resources such as virtual machines, virtual storage etc

→ Iaas is used by IT Admins.

→ Iaas provides only Infrastructure.

→ Example; AWS EC2, Azure VM etc.

2.a) Why do we need Autonomic Computing? Explain.

sol

Autonomic computing is a self-managing computing model to changing conditions without requiring extensive human oversight.
It aims to ~~enhance~~ reduce human intervention in managing complex systems.

We need Autonomic computing because of the following reasons:-

(i) Increasing Complexity of IT Systems:
→ Modern computing systems are highly complex. So, autonomic systems can manage themselves with minimal human intervention.

(ii) Reducing Operational costs:
→ Human maintenance and monitoring are highly expensive. So, autonomic computing automates tasks like updates, monitoring and optimization.

(iii) Minimizing Human Errors:
→ Human errors can cause downtime, data loss and security breaches. So, autonomic systems can self-correct, adapt and optimize automatically.

(iv) Improving System Reliability and Availability:
→ Autonomic computing can detect faults and recover without stopping the devices.

(v) Scalability and Performance:
→ Autonomic computing helps systems to scale dynamically based ~~de~~ on demand and also ensures optimal performance by tuning resources in real-time.

(vi) Security Management:
→ Autonomic computing can monitor threats, detect anomalies, and take preventive actions.

(vii) Self-configuration and ~~Aap~~ Adaptation:
→ Autonomic computing can configure themselves when integrated with new components.

Because of these reasons, autonomic computing is needed.

2.b) Differentiate between Cloud computing and Grid Computing.

sol

S.N.	Cloud Computing	Grid Computing.
i)	Cloud computing is a technology that uses remote servers on the internet to store, manage and access data online rather than local drivers.	Grid computing is a distributed architecture of multiple computer connected by networks to accomplish a joint task.
ii)	Cloud computing follows client server computing architecture.	Grid computing follows distributed computing architecture.
iii)	In cloud Computing, Resources are managed by service providers.	In grid computing, resources are managed by different organizations.
iv)	Cloud Computing provides services like IaaS, PaaS SaaS and more.	Grid computing provides only distributed computing tasks.
v)	Cloud computing are highly scalable.	~~Distri~~ Grid computing are limited scalable.
vi)	Cloud Computing is more flexible.	Grid Computing is less flexible.

S.N.	Cloud Computing	Grid Computing.
(vii)	Cloud computing operates as a centralized management system.	Grid computing operates as a decentralized management system.
(viii)	Cloud computing is service-oriented.	Grid computing is application oriented.
(ix)	Cloud computing is accessible through standard web protocols.	Grid computing is accessible through grid middleware
(x)	Cloud computing is used in web hosting, Saas apps, big data, IoT, AI etc.	Grid computing is used in scientific research, simulations, complex computation etc.

3.a) Explain hypervisor with its types.

sol → A hypervisor is also known as Virtual Machine Monitor (VMM).

→ A hypervisor is a software that enables the creation and management of Virtual Machines (VMs).

~~There are two ty~~

→ The main functions of hypervisor are as follows:-

(i) Virtualization of hardware resources

(ii) Isolation between virtual machines

(iii) Resource allocation and management

(iv) Monitoring and Control of virtual machines

→ There are two types of hypervisor:

(i) Type-1 Hypervisor (Bare-Metal Hypervisor):

→ Type-1 hypervisor is a native hypervisor that runs directly on the host's ~~operating~~ hardware to manage guest OS.

→ Type-1 hypervisor is used in data centers, servers and enterprise environments.

→ Examples: VMware ESXi, MS-Hyper-V, Xen, KVM (Kernel-based Virtual Machine).

→ The advantages of Type-1 hypervisor are as follows:

- Better Performance
- More secure and stable
- Ideal for large-scale environments

→ The diagram of Type-1 hypervisor are as follows:

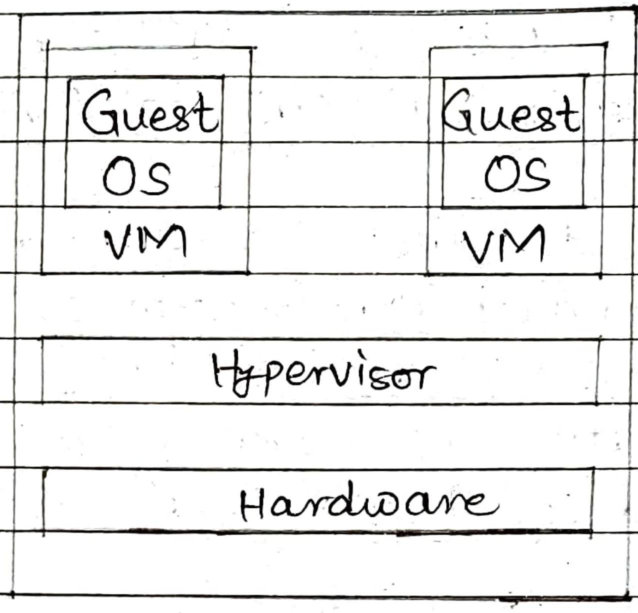

Fig. Type-1 Hypervisor

(ii) Type-2 Hypervisor (Hosted Hypervisor):

→ Type-2 hypervisor is a hosted hypervisor that runs on the top of a conventional OS and uses the host OS to access hardware resources.

→ Type-2 hypervisor is used for personal use, testing and development environments.

→ Examples:- Oracle VirtualBox, VMware Workstation, Parallels Desktop, etc.

→ The advantages of Type-2 hypervisor are as follows:-

- Easy to install and use
- Good for desktop virtualization.
- Suitable for beginners or testing.

→ The diagram of Type-2 hypervisor are as follows:-

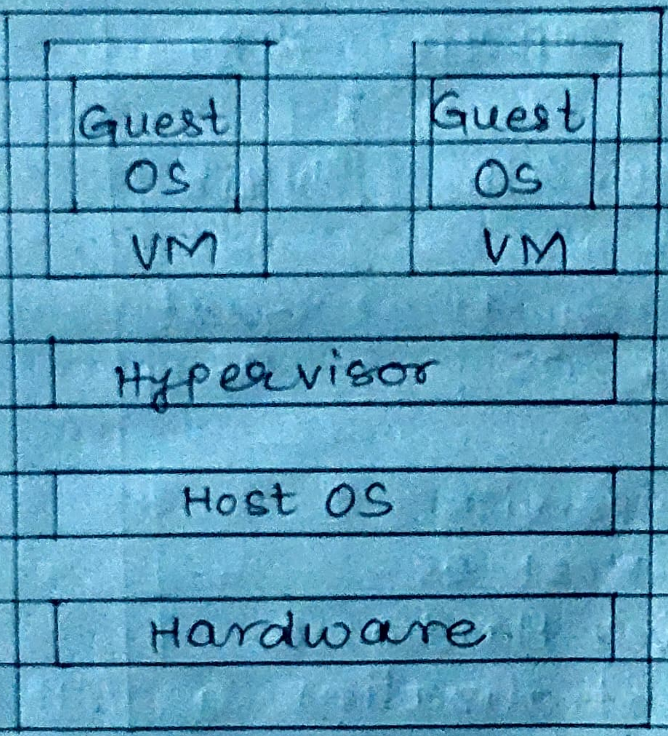

Fig. Type-2 Hypervisor

Q.3.b) Explain virtualization and VM with application.

sol (*) Virtualization:

Virtualization is the process of creating a virtual ~~machines~~ version of dedicated resources from a single hardware system.

The architecture of virtualization is shown below:

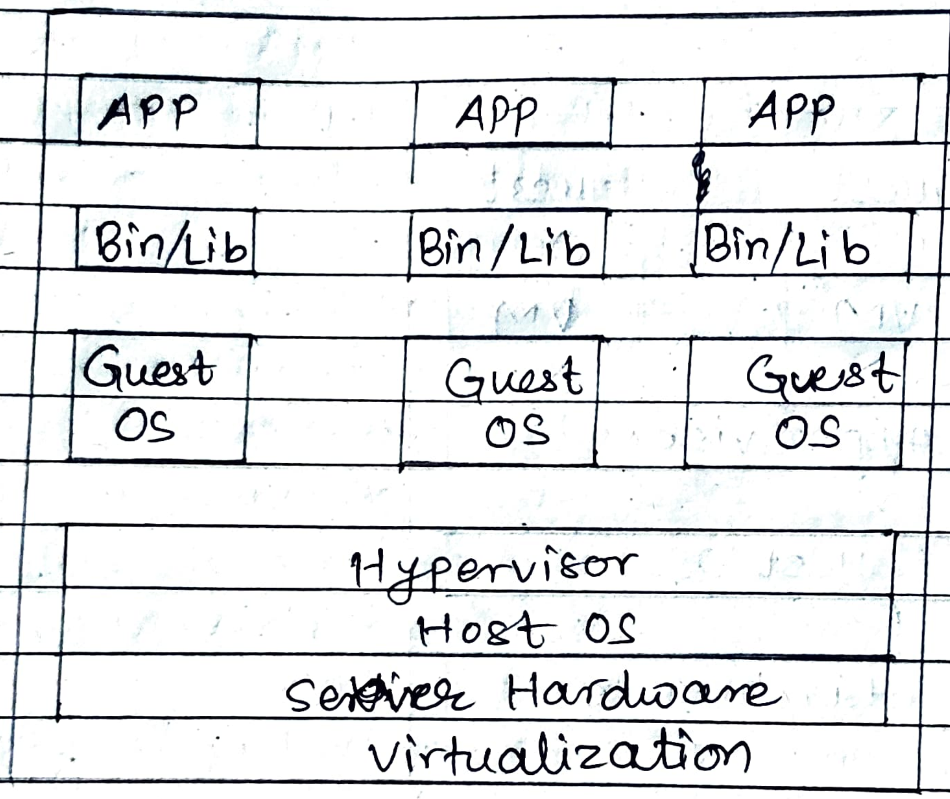

fig. Virtualization.

→ There are 6-types of virtualization in cloud computing. They are:

(i) Server Virtualization:
→ It divides a physical server into multiple virtual server.

(ii) Storage Virtualization:
→ It combines multiple physical storage devices into a single virtual storage pool.

(iii) Network Virtualization:
→ It creates virtual networks that operate independently of the physical network infrastructure.

(iv) Desktop Virtualization:
→ It enables users to access their desktop environment remotely.

(v) Application Virtualization:
→ It allows applications to run in environments separate from the underlying operating system.

(vi) Data Virtualization:
→ It brings data from different sources together in one place.

⊛ VM (Virtual Machines):

→ A virtual machine is a computer resource that uses software instead of a physical computer to run programs and deploy applications.

→ The ~~block diagr~~ architecture diagram of virtual machine is shown below:

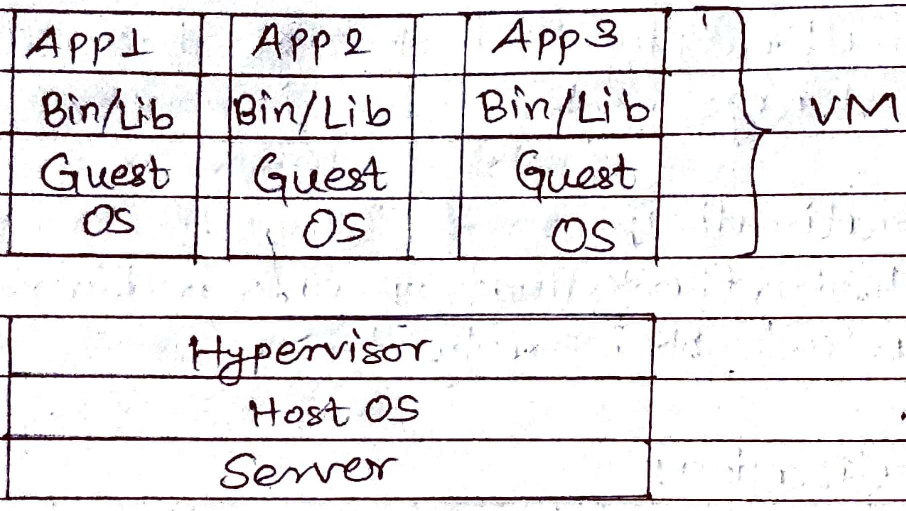

Fig. VM Architecture.

The working of virtual machines are as follows:-

(i) Initialization:

→ The hypervisor initiates and allocates resources to each VM upon startup.

(ii) Resource Management:

→ The hypervisor dynamically manages and distributes hardware resources among VMs based on demand and predefined policies.

(iii) Isolation:

→ Each VM is isolated from others i.e. one VM carsh does not affect others.

(iv) Execution:

→ Each VM runs its own OS and applications, independent of other VMs.

⇒ The applications of VMs are as follows:-

- Software Testing and Development
- Cyber Security and Malware Analysis
- Cloud Computing and server solidation
- Backup and Disaster Recovery
- Artificial Intelligence and Machine Learning
- Cross-Platform Capability

4. a) Explain which cloud ~~computing~~ deployment model is best for your outsourcing IT company.

Sol

The best cloud deployment model for an outsourcing IT company depends on the company's business goals, data sensitivity, scalability needs and client requirements.
However, the Hybrid cloud model is the most suitable for most outsourcing IT companies.

A hybrid cloud model combines two or more cloud deployment models typically a private cloud (for internal use) and a public cloud (for general).
It allows data and applications to move between them as needed.

Hybrid cloud model is best for outsourcing IT company because of the following reasons:

(i) Flexibility:
→ We can use both public and private cloud together based on needs.

(ii) Better Security:
→ We can keep sensitive data in the private cloud for more protection.

(iii) Cost effective:
→ We can use public cloud for regular work to save money and private cloud only when necessary.

(iv) Handles Many Clients:
→ Hybrid model can support both types of clients who want strong privacy and who don't.

(v) Scalable:
→ We can easily increase or decrease our cloud resources during big or small projects.

(vi) Fast Backup and Recovery:
→ If something goes wrong, data or apps can quickly move to another cloud system.

(vii) Improves Performance:
→ We can ~~run~~ different parts of our applications in the most suitable cloud for better speed.

(viii) Supports Remote Work:
→ Employees can access system from anywhere through the public cloud, which secure work stays in the private cloud.

4.b) Compare different kinds of clouds available;

Sol :-

Features	Public cloud	Private cloud	Hybrid Cloud	Community Cloud
Ownership	Third-Party	Single organization	Mixed	Group / Organization
Access Level	Public	Private	Both	Shared only
Cost	Low	High	Medium	Medium
Security	Medium	High	High	High
Scalability	High	Medium	High	Medium

5.a) Explain Map Reduce with its architecture and Working.

Sol :- MapReduce is the programming model that allows user to process huge data in distributed Hadoop system.

MapReduce provides platforme to perform distributed and parallel processing on large data sets in a distributed environment.

The architecture of MapReduce is shown below:

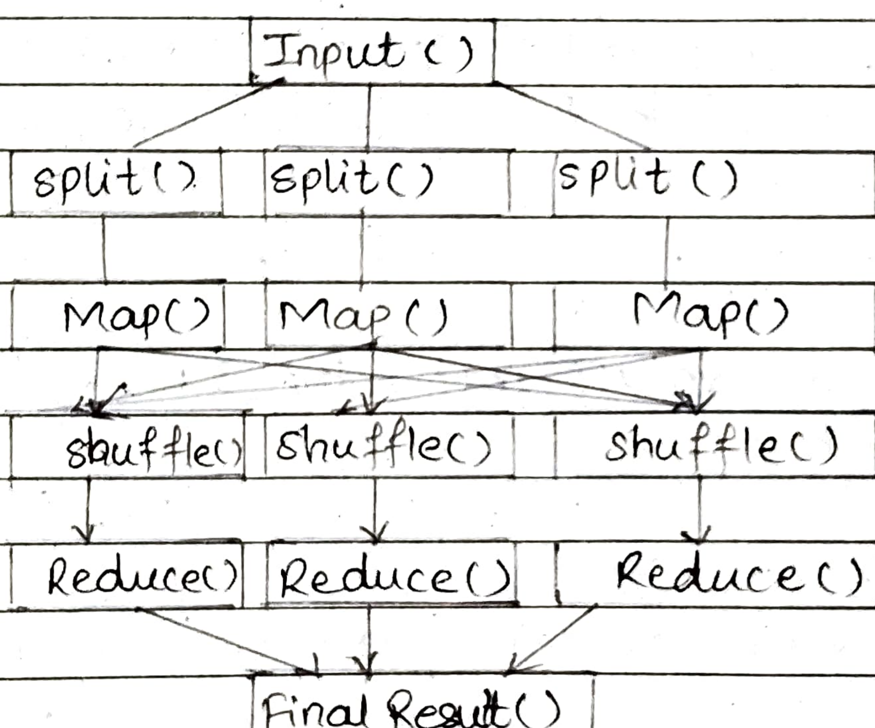

Fig. MapReduce Architecture

The working of MapReduce are as follows:-

(i) Input():
→ Input are the raw text that we want to analyze.

(ii) Split():
→ Splitting is the process of dividing the input data into chunks.

(iii) Map():
→ In mapping, each chunk is processed by a Map function that converts the text into key value pairs.

(iv) Shuffle():
→ In shuffling, all values with the same key are grouped together.

(v) ~~Reducing~~ Reduce():
→ In reducing, each group is processed to sum the values.

(vi) Final Result():
→ In final result, all word counts are collected together.

5.b) List out the guiding principles and advantages of SOA.

Sol

→ SOA stands for Service-Oriented Architecture.

→ SOA is an architectural approach in which applications make use of services available in the network.

The SOA stack can be categorized into two parts— functional aspects and quality of service aspects.

Functions

- Service Registry
- Business Process
- service
- Service Description
- Service Communication Protocol
- Transport

Quality of Service

- Policy
- Security
- Transaction
- Management

(*) Functional aspects:

- Transport:

→ It transports the service requests from the service consumer to the service provider and service responses from the service provider to the service consumer.

- Service Communication Protocol:

→ It allows the service provider and the service consumer to communicate with eachother.

- Service Description:

→ It describes the service and data required to invoke it.

- Service:

→ It is an actual service.

- Business Process:

→ It represents the group of services called in a particular sequence associated with the particular rules to meet the business requirements.

- Service Registry:

→ It contains the description of data which is used by service providers to publish their services.

(*) Quality of Service Aspects:

- Policy:

→ It represents the set of protocols according to which a service provider make and provide the services to consumers.

- Security:

→ It represents the set of protocols required for identification and authorization.

- Transaction:

→ It provides the surety of consistent results.

- Management: ~~defines~~

→ It defines the set of attributes used to manage the services.

(*) Guiding Principles of SOA:

(i) Loose Coupling:

→ Services are independent of each other.

(ii) Reusability:

→ Services are designed to be used in multiple applications or processes.

(iii) Interoperability:

→ Services should work across different platforms and technologies.

(iv) Discoverability:

→ Services should be easy to find using a service registry.

(v) Abstraction:

→ Service consumers only know what the service does, not how it does.

(vi) Composability:

→ Small services can be combined to build larger, more complex applications.

* Advantages of SOA:

- Reusability:

→ Services can be reused in different applications, saving time and effort.

- Easy Maintenance:

→ Individuals services can be updated or replaced without affecting the whole system.

- Faster Development:

→ Services speeds up new application development

- Scalability:

→ Individual services ~~can~~ scale easily based on demand.

- Cost Efficiency:

→ Reuse of services reduces development and maintenance cost.

- Flexibility:

→ Services can be built using any technology and still work together.

6.a) Explain cloud security challenges?

Sol:

The cloud security challenges are as follows:-

(i) Data Breach:

→ Hackers may steal private data like passwords, files or consumers' information.

(ii) Data Loss:

→ Important data may get deleted or lost due to mistakes, attacks or system failure.

(iii) Account Hijacking:

→ Attackers gain access to a user's cloud account by phishing or weak passwords.

(iv) Insecure APIs:

→ APIs are used to interact with cloud services. If not properly secured, hackers can exploit them to gain access.

(v) Insider Threat:

→ Employees or cloud staff may misuse their access to harm or steal data.

(vi) Less Control Over Data:
→ In cloud, data is stored on someone ~~elee~~ else's servers like Google or Amazon.

(vii) Shared Resources:
→ In public cloud, many users share the same system. If one system has a weakness, others may be affected too.

(viii) Compilance and Legal Issues:
→ Data may be stored in another country. It must follow local laws or you could face legal trouble.

6.b) How do you handle Disaster in Cloud?

Sol

A disaster in cloud means a major failure like data loss, system crash, cyber attacks or natural disaster that affects cloud services.

We handle disasters in cloud by following ways:-

(i) Regular Backups:-

→ Regular backups helps restore files quickly if they are lost or corrupted.

(ii) Multiple Data Centers:

→ If one server or area fails, another can take over.

(iii) Disaster Recovery as a Service (DRaaS):

→ Cloud providers offer DRaas to ~~help~~ recover apps and data quickly

(iv) Automatic Failover Systems:

→ If one server or system goes down, traffic automatically moves to a working system.

(v) Data Replication:
→ Real time copying of data to another cloud server, ensures the latest data is always available.

(vi) Testing and simulation:
→ Regularly test disaster recovery plans using simulations to find problems before a real time disaster happens.

(vii) Access Control and Security:
→ Prevent unauthorized users from accessing or damaging systems during a disaster.

(viii) Disaster Recovery Plan (DR plan):
→ When a disaster occurs, follow a written step-by-step guide on what to do.

7. Write short notes on:

a) Hadoop:

sol

→ Hadoop is an open-source framework used for storing and processing large datasets across multiple computers.

→ It uses the MapReduce programming model to process data in parallel.

→ It is used in big data analytics, log analysis, recommendation systems.

→ The Components of Hadoop are as follows:-

- HDFS (Hadoop Distributed File system):

→ It ~~uses~~ stores data by splitting it into blocks and distributing it across machines.

- MapReduce:

→ It processes data into two steps - Mapping & Reducing

- YARN:

→ It manages resources and job scheduling.

- Hive/ Pig:

→ It is a tool ~~for~~ querying and analyzing data easily.

(b) Vulnerability Assessment:

Sol

→ Vulnerability assessment is the process of identifying, analyzing and classifying security weakness in systems, networks or software.

→ It has goal to prevent attacks by fixing security gaps before hackers can exploit them.

→ The steps involved in vulnerability assessment are as follows:-

(1). Scanning systems using tools like Nessus or OpenVAS.

(2). Finding weaknesses such as outdated software or weak passwords.

(3). Analyzing risks based on how likely and harmful they are.

(4). Reporting with recommendations to fix them.

(c) Microsoft Azure Cloud:

sol:

→ Microsoft Azure is a cloud computing platform by Microsoft that offers over 200 services including virtual machines, databases, storage and AI tools.

→ It is used for apps hosting, data storing, AI models running and building web services.

→ The main features of MS-Azure are as follows:-

(i) It supports IaaS, PaaS and SaaS.

(ii) It provides services like Azure VMs, Azure SQL Database, Azure Functions.

(iii) It is highly available for global network of data centers.

(iv) It is welly integrated with Windows and Microsoft 365.